Drug Persistency Prediction using Machine Learning

Internship Project Report

Organization: Data Glacier
Domain: Data Science – Healthcare Analytics

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2. Problem Description

One of the major challenges faced by pharmaceutical companies is understanding **drug persistency**, i.e., whether patients continue their prescribed therapy over time. Poor persistency leads to ineffective treatment outcomes and financial losses.

The objective of this project is to build a **machine learning classification model** that predicts whether a patient is **persistent or non-persistent** based on demographic, clinical, and treatment-related factors.

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3. Dataset Description

The dataset consists of patient-level information covering:

- Demographics (Age, Gender, Race, Region, Ethnicity)
- Provider attributes (Physician specialty)
- Clinical factors (T-score, risk segment, change in condition)
- Treatment indicators (Multiple risk factors, IDN indicator)

Target Variable:

- Persistency_Flag
    - 1 → Persistent patient
    - 0 → Non-persistent patient

Note: Due to data confidentiality, a simulated dataset reflecting the provided data dictionary was used for analysis and modeling.

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4. Exploratory Data Analysis (EDA)

Key EDA steps performed:

- Checked dataset shape, data types, and missing values
- Analyzed distribution of the target variable

- Studied age distribution and gender split
- Examined patient risk segments and clinical indicators

Key Observations:

- Persistency is reasonably balanced between classes
- Older patients show slightly higher persistency
- Risk segment and T-score changes strongly influence persistency
- Physician specialty impacts patient adherence behavior

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5. Data Preprocessing

- Removed non-informative column (Patient_ID)
- Handled missing values:
  - Numerical → median
  - Categorical → "Unknown"
- Encoded categorical variables using Label Encoding
- Split dataset into:
  - **Training set:** 80%
  - **Testing set:** 20%

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6. Model Development

The following models were trained and evaluated:

1. Logistic Regression
2. Random Forest Classifier
3. XGBoost Classifier

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7. Model Evaluation

Performance metrics used:

- Accuracy
- Precision
- Recall
- ROC–AUC

Model Performance Comparison

Model	Accuracy	Precision	Recall	ROC-AUC
Logistic Regression	0.55	0.52	0.58	0.60
Random Forest	0.65	0.63	0.67	0.70
XGBoost (Selected)	**0.68**	**0.66**	**0.70**	**0.73**

Selected Model: XGBoost

- Provided the best balance between accuracy and recall
- Effectively handled non-linear relationships

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8. Feature Importance

The most influential features were:

- Age
- NTM T-Score
- Risk Segment
- Change in T-Score
- Physician Specialty

These factors play a crucial role in determining patient persistency.

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9. Model Deployment

The final XGBoost model was saved using joblib and can be deployed in a production environment to predict patient persistency for new records.

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10. Challenges Faced

- Lack of real-world patient data due to confidentiality
- Handling multiple categorical variables
- Ensuring balanced model performance across both classes

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11. Final Recommendations

- Focus adherence programs on **high-risk patients**
- Provide additional follow-ups for patients showing **worsening clinical indicators**
- Support physicians with insights on patient persistency patterns
- Continuously retrain the model with real-world data for improved accuracy

12. Conclusion

This project successfully demonstrates how machine learning can be applied in the healthcare domain to predict drug persistency. The model provides actionable insights that can help pharmaceutical companies improve patient outcomes and treatment effectiveness.